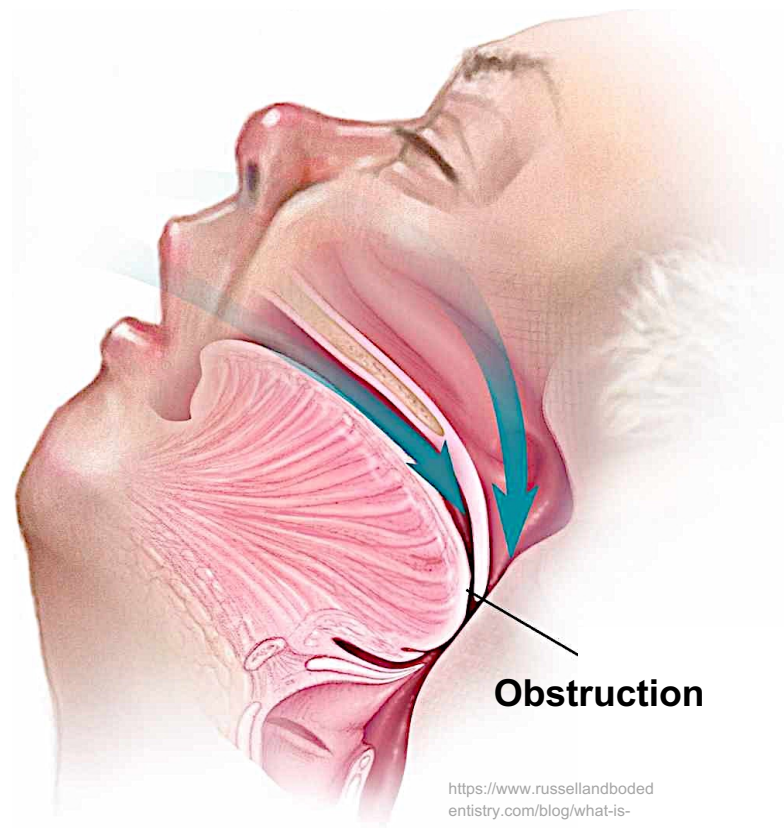


Hydrogel Injection to Stiffen the Soft Palate in OSA*

*Obstructive Sleep Apnea

Group 8.1

Elsässer Léon, Jourdan Lucas, Vuagniaux Aurélie, Zell Kilian



<https://www.russellandbodedentistry.com/blog/what-is-obstructive-sleep-apnea/>

Table of Contents

- **What is OSA?**
- Problem & Our Mission
- Proposed Solution
- Results & Discussion



Obstructive Sleep Apnea

- Repeated blockage of the upper airway during sleep
- Disrupts the sleep-wake cycle: **Fragmented sleep**

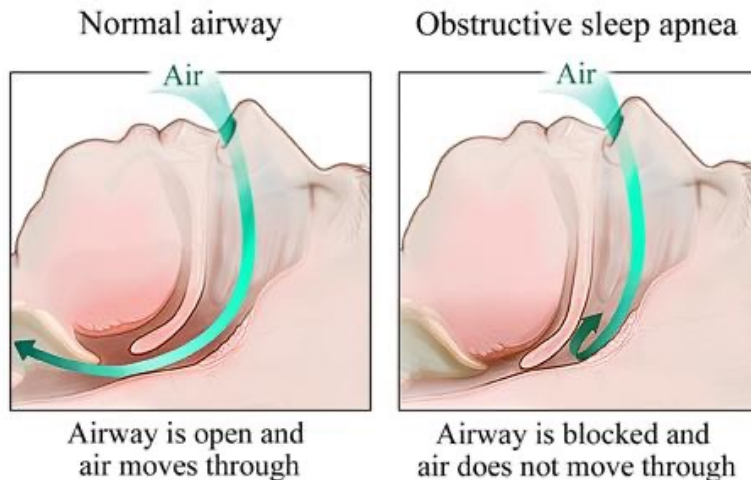
→ **Low sleep quality, daytime sleepiness and long-term impacts on health**

Obstructive Sleep Apnea Stats

Number of people affected

400'000

Swiss Federal Statistical Office.
(n.d.). Effectif et évolution de la population selon l'âge. Retrieved from
<https://www.bfs.admin.ch/bfs/fr/home/statistiques/population/effectif-evolution/age.assetdetail.26565155.html>



University of Michigan Health. (n.d.). Obstructive sleep apnea. Michigan Medicine. Retrieved May 21, 2024, from <https://www.uofmhealth.org/conditions-treatments/brain-neurological-conditions/obstructive-sleep-apnea/physicians>

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Problem



Limitations of Current Treatments



Our Mission

Dr. Karma Lambercy
Médecin associé
Responsable unité SAOS
CHUV



Goal

“Design a surgical procedure to
stiffen the oropharyngeal tissues ...



Idea

...using an injectable hydrogels*.”

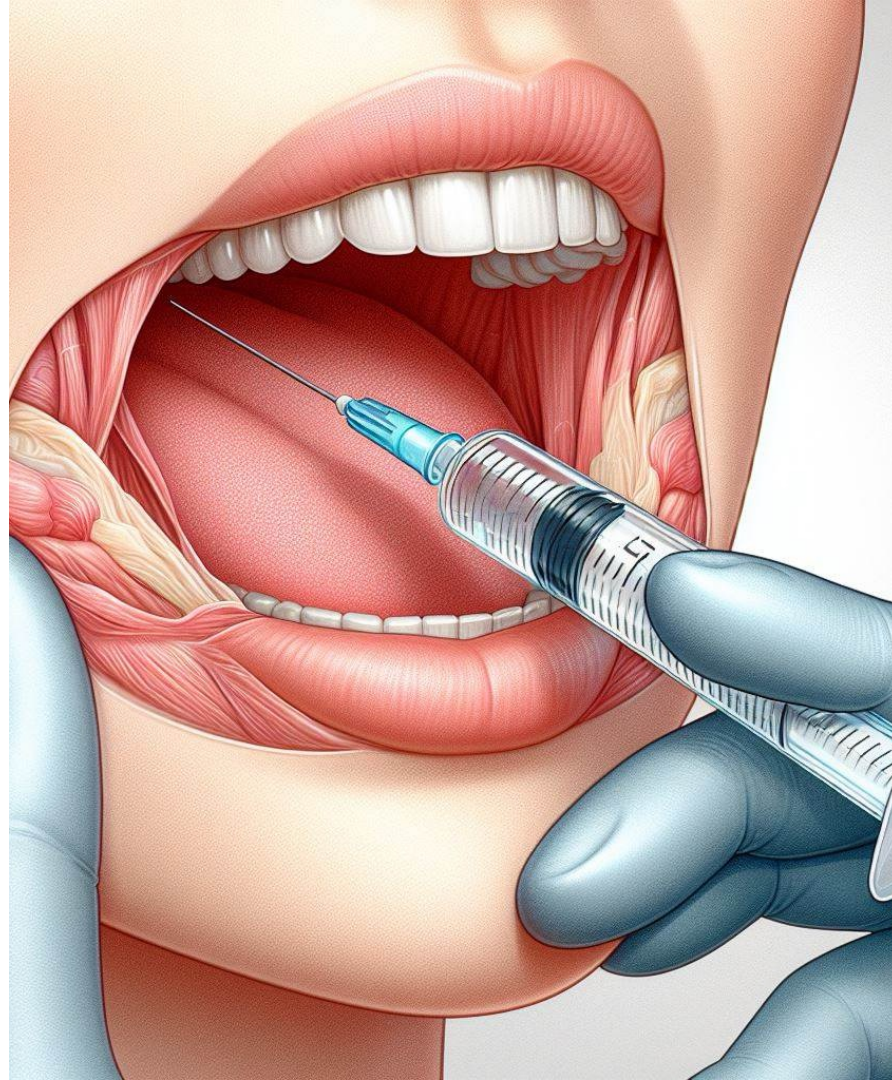


* Jelly-like material made of a 3D network of polymers that can hold a large amount of water.

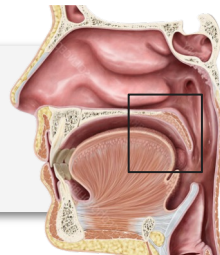
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Image generated by DALL-E 3, with prompt: "Generate a clear, realistic medical illustration showing an injection procedure on the soft palate."



Hydrogel Injection in the Soft Palate



What ?

- Injection of a stiffening hydrogel in the soft palate
- Between fat and muscle tissues



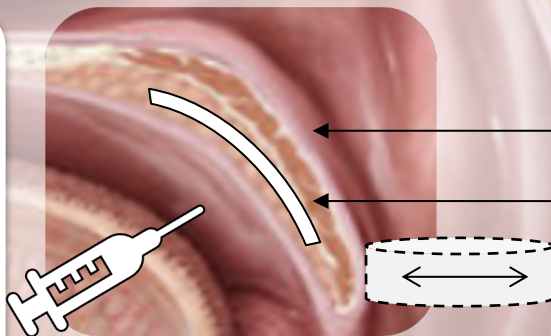
Why ?

- Add internal mechanical support
- Reduce displacement
- Eliminate obstruction



How ?

- Multi-injection in liquid state
- Injection through the oral cavity
- In-office, topical anesthetic spray



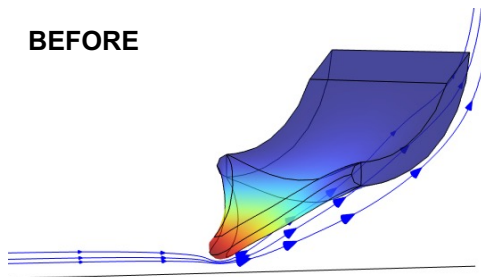
Adapted from: <https://www.alamy.com/this-image-shows-the-upper-organs-of-the-respiratory-system-which-image156174539.html?imageid=3D4AB703-60E7-4268-8247-63C29AB642A4&p=532318&pn=1&searchid=27e1eccfbfb77eca08e46b28e52dbb35&searchtype=0>

Soft Palate

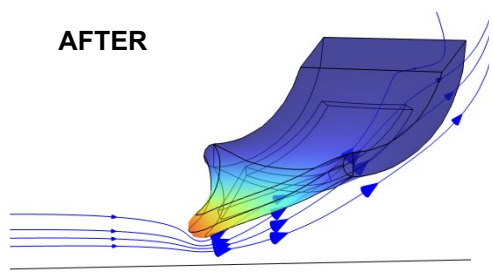
Hydrogel Injection

Obstruction Zone

BEFORE



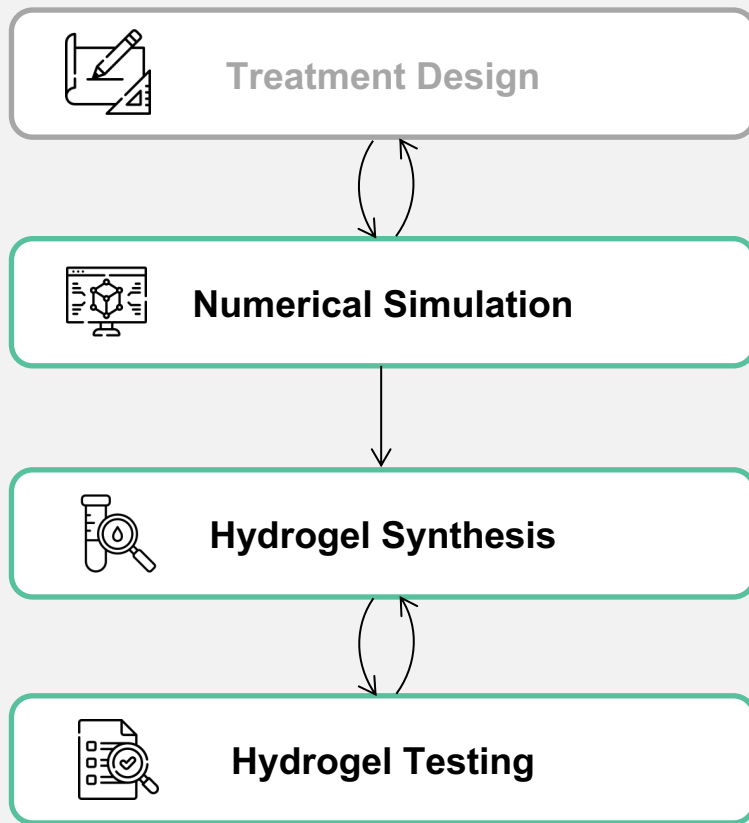
AFTER



Expected result after treatment, modeled during maximal expiration rate, based on COMSOL numerical simulation

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Numerical Simulation

- Patient modelled in supine* position. *Laying on the back

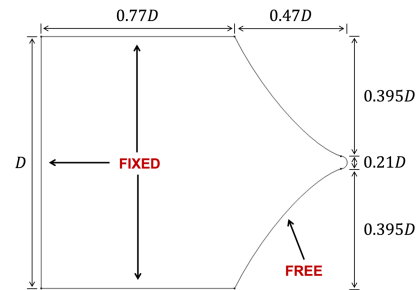
- Simplified geometry based on dimensions from the literature.

→ Fully coupled Solid/Fluid (turbulent flow, $k-\epsilon$ -model)

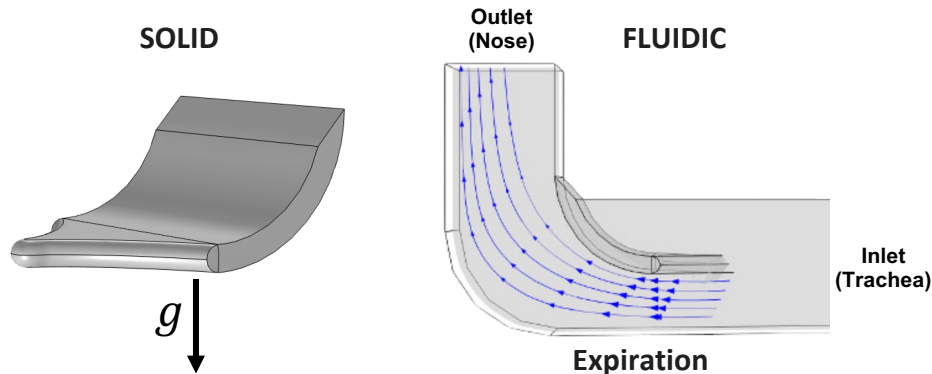
Material Properties Used

	Palate	Hydrogel
E [Pa]	6 500	60 000
ν [-]	0.45	0.49
ρ [kg/m ³]	1060	1000

Dimensions of the Soft Palate



Two Experimental Settings

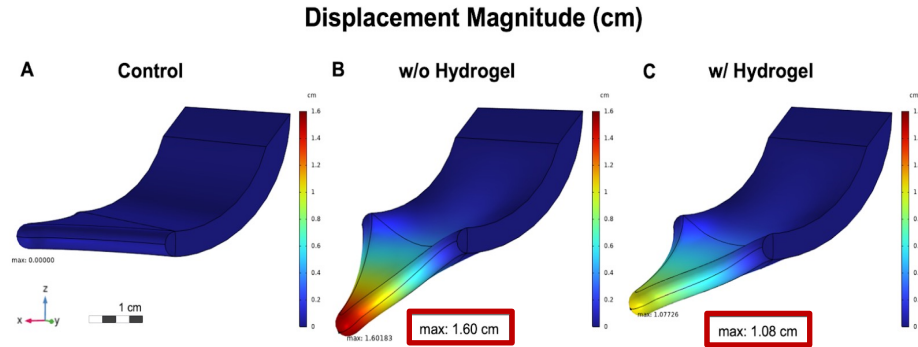


Results

Consideration

- Elastic attachment of the hydrogel
- Expiration at maximum flow rate ($800 \text{ cm}^3/\text{s}$)
- Outlet pressure 1 bar

SOLID

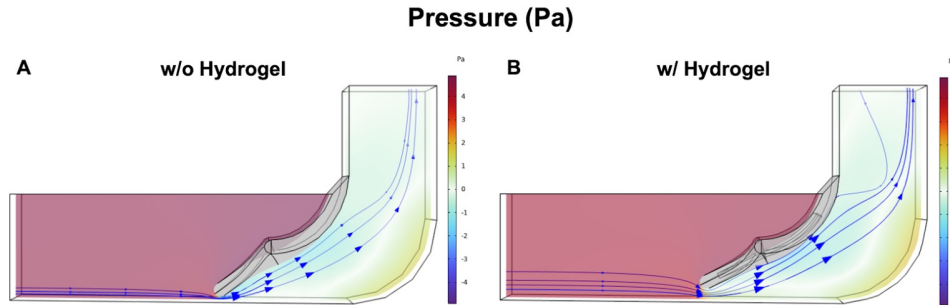


w/ Hydrogel vs w/o Hydrogel

-0.5cm

Maximal Displacement

FLUIDIC

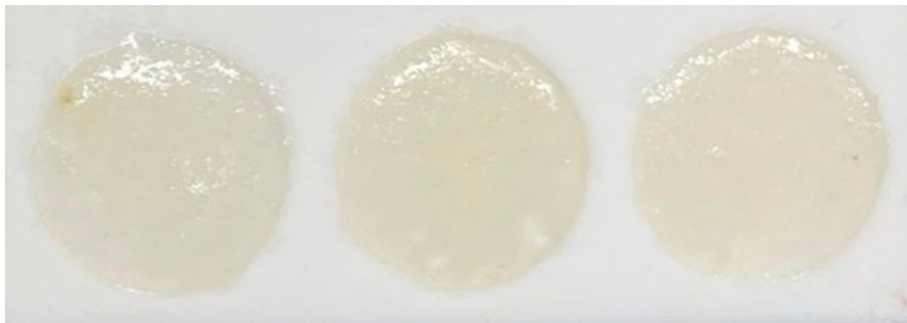


-10% Pressure Loss

-14% Average Stress

Synthesis hydrogel

- Composition is based on a C/GP solution with the addition of alginate and CaCl_2
 - ➔ Strength is developed thanks to ionic bonds
- Gelation method is a chemical reaction

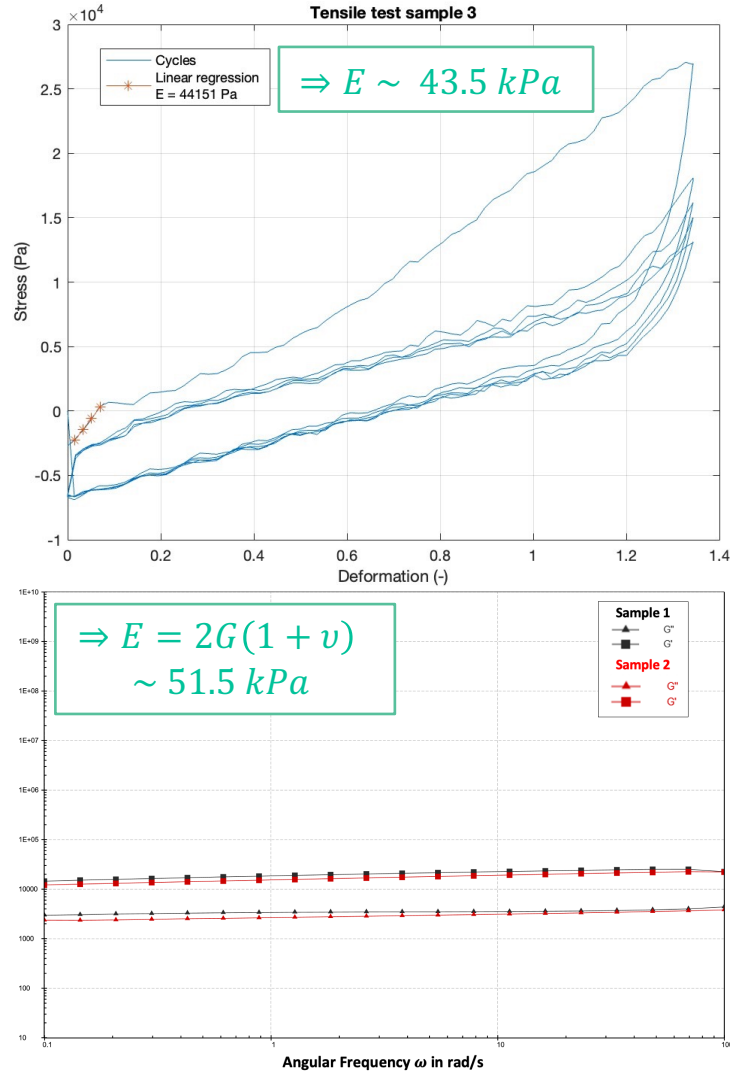


Objectives:

- Higher strength (4 to 5 times) than soft tissues
- Simple formulation
- Injectability
- Biocompatibility
- **Avoid UV curing inside human tissues**
- (Thermo – responsiveness)

Mechanical properties

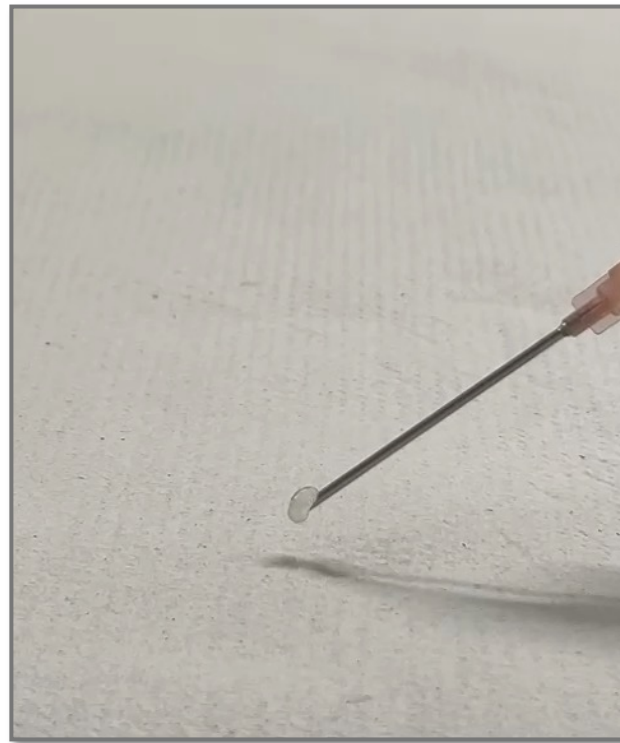
- Tensile: 100 *mm/min*
- Rheological: Linear viscoelastic region
- 37°C
- ➔ Suitable Young's Modulus
- ➔ Convenient cyclic behavior evaluation



Injectability proof

CHUV surgery equipment

- Double canal syringe to start the chemical reaction once injected



Thank You !

Any questions?

<https://emedicine.medscape.com/article/295807>
<https://www.russellandbodedentistry.com/blog/what-is-obstructive-sleep-apnea/>

Normal



Obstructive Sleep Apnea

